Practical No. 3– Circular Queue using Array

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Problem Statement: Circular Queue-- CO1, CO2

Implement Circular Queue using Array.

Perform following operations on it.

a. Insertion (Enqueue)

b. Deletion (Dequeue)

c. Display

Code:

```
#include<iostream>
using namespace std;
#define SIZE 5 class
CircularQueue
{
int queue[SIZE]; int
front, rear; public:
CircularQueue()
{
front = -1; rear
= -1;
}
void enqueue()
{
int value; if((front == (rear +
1) % SIZE))
{
```

```
cout << "\nQueue Overflow"; return;
}
cout << "\nEnter Element : ";
cin >> value; if(front == -1) {
front = rear = 0;
}
else
{
rear = (rear + 1) % SIZE;
}
queue[rear] = value; cout << "\nElement
Inserted Successfully.";
}
void dequeue()
{
if(front == -1)
{
cout << "\nQueue Underflow"; return;
}
cout << "\nDeleted Element : " << queue[front]; if(front
== rear)
{
front = rear = -1;
}
else { front = (front + 1)
% SIZE;
```

```
}
}
void display()
{
if(front == -1)
{
cout << "\nQueue is Empty"; return;
}
cout << "\nQueue Elements : ";
int i = front; while(true)
{
cout << queue[i] << " ";
if(i == rear) break; i = (i
+ 1) % SIZE;
}
cout << endl;
} };
int main() { CircularQueue q;
int choice; do
{
cout << "\n\n===== CIRCULAR QUEUE =====";
cout << "\n1. Enqueue"; cout << "\n2.
Dequeue"; cout << "\n3. Display"; cout <<
"\n4. Exit"; cout << "\n\nEnter Choice : "; cin
>> choice; switch(choice)
{
```

```
case 1:
q.enqueue();
break; case
2:
q.dequeue();
break; case
3:
q.display(); break; case 4: cout <<
"\nProgram Terminated"; break;
default: cout << "\nInvalid Choice";
}
}while(choice != 4); return
0;
}
```

Output:

```
==== CIRCULAR QUEUE =====1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice : 1 Enter Element
:  50  40  30  20  10  Element
Inserted Successfully.
===== CIRCULAR QUEUE =====
1. Enqueue
2. Dequeue
3. Display
```

4. Exit

Enter Choice : 2

Deleted Element : 50

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display 4. Exit

Enter Choice : 1 Enter

Element : 60

Element Inserted Successfully.

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 3

Queue Elements : 60

===== CIRCULAR QUEUE =====

1. Enqueue

2. Dequeue

3. Display

4. Exit

Enter Choice : 4

Program Terminated